

Algebra 1
Unit 1
Lesson 9 Homework

Name **Answer Key** _____
Date _____
Period _____

1. Which of the following expressions can be combined?

- A. $-3\sqrt{25} - 25\sqrt{3}$ $2\sqrt{4 \cdot 3} - 6\sqrt{9 \cdot 3}$
 B. $\sqrt{7} + 4\sqrt{14}$ $2 \cdot 2\sqrt{3} - 6 \cdot 3\sqrt{3}$
 C. $2\sqrt{12} - 6\sqrt{27}$ $4\sqrt{3} - 18\sqrt{3}$
 D. $5\sqrt{25} + \sqrt{20}$ $-14\sqrt{3}$

Three students simplified the expression $6\sqrt{27} - 5\sqrt{12}$ below. Their work is shown below.

Marc	Jess	Tamara
$6\sqrt{27} - 5\sqrt{12}$ $1\sqrt{13}$ $\sqrt{13}$	$6\sqrt{27} - 5\sqrt{12}$ $6\sqrt{9 \cdot 3} - 5\sqrt{4 \cdot 3}$ $54\sqrt{3} - 20\sqrt{3}$ $34\sqrt{3}$	$6\sqrt{27} - 5\sqrt{12}$ $6\sqrt{9 \cdot 3} - 5\sqrt{4 \cdot 3}$ $18\sqrt{3} - 10\sqrt{3}$ $8\sqrt{3}$

2. Which student simplified the expression correctly?

Tamara simplified the expression correctly.

3. Write a note to one of the students who made an error explaining their mistake.

Answers will vary.

Mark; first, in order to combine radical expressions they must be like terms, and these expressions are not. Secondly, you cannot add radicands together due to the order of operations.

4. Write a note to the other student who made an error explaining their mistake.

Answers will vary.

Jess, you did the first step correctly. However, when you simplified the radicands you forgot to take the principal square root of the numbers.

For question 5-10, find the sum or difference of the expression. If you cannot add or subtract the expression, explain why.

$$\begin{aligned} 5. \quad & -2\sqrt{18} + \sqrt{50} \\ & -2\sqrt{9 \cdot 2} + \sqrt{25 \cdot 2} \\ & -2 \cdot 3\sqrt{2} + 5\sqrt{2} \\ & -6\sqrt{2} + 5\sqrt{2} \\ & \boxed{-\sqrt{2}} \end{aligned}$$

$$\begin{aligned} 6. \quad & \sqrt[3]{56} - \sqrt[3]{189} \\ & \sqrt[3]{8 \cdot 7} - \sqrt[3]{27 \cdot 7} \\ & 2\sqrt[3]{7} - 3\sqrt[3]{7} \\ & \boxed{-\sqrt[3]{7}} \end{aligned}$$

$$\begin{aligned} 7. \quad & 6\sqrt{8} - 8\sqrt{6} \\ & 6\sqrt{4 \cdot 2} - 8\sqrt{6} \\ & 6 \cdot 2\sqrt{2} - 8\sqrt{6} \\ & 12\sqrt{2} - 8\sqrt{6} \end{aligned}$$

These cannot be combined because they are unlike terms.

$$\begin{aligned} 8. \quad & 200^{\frac{1}{2}} - 1000^{\frac{1}{3}} \\ & \sqrt[2]{100 \cdot 2} - \sqrt[3]{1000} \\ & 10\sqrt{2} - 10 \end{aligned}$$

These cannot be combined because they are not like terms.

$$\begin{aligned} 9. \quad & -\sqrt{80} + 6\sqrt{5} - 3\sqrt{20} \\ & -\sqrt{16 \cdot 5} + 6\sqrt{5} - 3\sqrt{4 \cdot 5} \\ & -4\sqrt{5} + 6\sqrt{5} - 3 \cdot 2\sqrt{5} \\ & 2\sqrt{5} - 6\sqrt{5} \\ & \boxed{-4\sqrt{5}} \end{aligned}$$

$$\begin{aligned} 10. \quad & \sqrt[3]{250} - 4\sqrt[3]{54} \\ & \sqrt[3]{125 \cdot 2} - 4\sqrt[3]{27 \cdot 2} \\ & 5\sqrt[3]{2} - 4 \cdot 3\sqrt[3]{2} \\ & 5\sqrt[3]{2} - 12\sqrt[3]{2} \\ & \boxed{-7\sqrt[3]{2}} \end{aligned}$$